Tau-Stress Zones: A Falsifiable Environmental–Neurodynamics Hypothesis for State-Dependent Anomalous Reports

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Abstract

We propose and operationalize the *Tau-Stress Zone* (TSZ) hypothesis: co-occurring environmental low-frequency drivers—extremely low frequency electromagnetic fluctuations (ELF), infrasound (INF), and slow geomagnetic variability—interacting with internal arousal can bias cortical network dynamics toward altered stability, captured by a *dimensionless* proxy τ. TSZ is a *state-dependent* mechanism intended to explain increases in ambiguous or false-positive perception/reporting under specific conditions without invoking exotic physics. We provide deterministic synthetic evidence using: (i) environmental surrogate streams with rare co-peaks; (ii) a ring-coupled Wilson–Cowan network perturbed by weak, jittered, sub-threshold drives under low vs. high arousal; (iii) a signal-detection mapping linking τ to false positives; and (iv) a spatial proximity prediction near infrastructure. Figures are robust *figure-tables* (no external graphics) with *exact* numbers from a deterministic seed. In this run, high+INF slightly *increased* τ relative to baseline, highlighting that the *direction* of effects is parameter-sensitive and must be empirically constrained. The hypothesis remains falsifiable by co-occurrence analyses, band-specific tests, and spatial gradients collected in preregistered field studies.

1 ENT in brief (self-contained)

Environmental Neurodynamics of **Tau-systems** (ENT) is a structural language: a **τ-system** is any dynamical regime whose effective stability against perturbations is summarized by a *dimensionless* quantity τ. A **Tau-Stress Zone** (TSZ) is a spatiotemporal window where exogenous drivers (ELF, INF, slow geomagnetic variability) and internal state (arousal) jointly depress or otherwise alter τ, changing perceptual decision thresholds under ambiguity. ENT assumes no extra dimensions or metaphysical entities; it is purpose-built to generate falsifiable predictions about state-dependent human perception and reporting.

2 Falsifiable hypothesis and predictions

TSZ Hypothesis. When ELF (20–30 Hz), INF (10–20 Hz), and slow geomagnetic variability *co-peak* during *high arousal*, a cortical network is more likely to enter transient altered-τ windows that (under ambiguous conditions) shift criteria and increase false-positive reports.
Predictions.

1. **Co-occurrence:** Windows with co-peaks in ($\mathrm{ELF}_{20\text{–}30}$, $\mathrm{INF}_{10\text{–}20}$, slow GM) show higher ambiguous-report odds than matched non-coincidence windows.

2. **Band specificity:** Out-of-band features weaken effects.

3. **Spatial gradient:** A peaked report gradient appears near infrastructure where residual ELF variance overlaps an INF hotspot.

4. **Dose irrelevance:** Ionizing dose at public perimeters is negligible and does not explain TSZ; TSZ is a dynamics/variance hypothesis, not a dose hypothesis.

3 Methods

3.1 Tier 1: Environmental surrogates

We synthesize 10-minute streams at 100 Hz: ELF (1–100 Hz; $1/f$ + bursts near 25 Hz), INF (0.5–20 Hz; machinery-like bursts at 10–20 Hz with slow AM), and a slow "geomagnetic" surrogate (very-low-frequency drift). In 60 s windows (10 s stride) we compute:

$$\mathrm{ELF}_{20\text{–}30} = \text{band power}(20\text{–}30\,\mathrm{Hz}), \quad \mathrm{INF}_{10\text{–}20} = \text{band power}(10\text{–}20\,\mathrm{Hz}), \quad \mathrm{GM} = \text{window mean},$$

then standardize each feature and define the coincidence score $C = z(\mathrm{ELF}_{20\text{–}30}) + z(\mathrm{INF}_{10\text{–}20}) + z(\mathrm{GM})$.

3.2 Tier 2: Neural dynamics and τ-proxy

We simulate a 20-node ring-coupled Wilson–Cowan network at 200 Hz, 60 s/condition. For node i,

$$\tau_e \dot{E}_i = -E_i + S\big(w_{ee}E_i - w_{ei}I_i + P_e + u_{e,i}(t)\big) + \kappa \sum_{j\in\mathcal{N}(i)} (E_j - E_i), \tag{1}$$

$$\tau_i \dot{I}_i = -I_i + S\big(w_{ie}E_i - w_{ii}I_i + P_i + u_{i,i}(t)\big) + \kappa \sum_{j\in\mathcal{N}(i)} (I_j - I_i), \tag{2}$$

with logistic $S(x) = (1 + \mathrm{e}^{-a(x-\theta)})^{-1}$ and $\mathcal{N}(i)$ the two ring neighbors. **Arousal** is modeled as increased background noise and a shallower a. Weak, jittered drives emulate **ELF** ($\approx$25 Hz, primarily into I) and **INF** ($\approx$15 Hz, modest to E/I). We study five conditions: A (low arousal), B (high arousal), C (high+ELF), D (high+INF), E (high+ELF+INF).

Stability proxy τ (dimensionless). Band-pass each $E_i(t)$ to 20–30 Hz, extract phases $\phi_i(t)$, and compute the Kuramoto order parameter

$$R(t) = \left| \frac{1}{N} \sum_{i=1}^{N} \mathrm{e}^{\mathrm{i}\phi_i(t)} \right| \in [0, 1].$$

For the mean excitatory signal $\overline{E}(t) = \frac{1}{N}\sum_{i=1}^{N} E_i(t)$, compute normalized spectral entropy H over 1–20 Hz. Over sliding 2 s windows (0.5 s step),

$$\tau \;\propto\; \frac{R}{(1-R) + 0.5\,H}.$$

We summarize by condition and report the fraction of windows below the *baseline* (A) 10th percentile.

Simulation constants (deterministic).

Seed	`424242` (deterministic)
Duration & fs	60 s/condition at 200 Hz (Tier 2); 10 min at 100 Hz (Tier 1)
Windows / stride	2 s / 0.5 s (Tier 2); 60 s / 10 s (Tier 1)
Nodes / topology	20 (ring), neighbors $\mathcal{N}(i) = \{i \pm 1\}$, coupling κ moderate
Nonlinearity	$S(x) = (1 + \mathrm{e}^{-a(x-\theta)})^{-1}$; high arousal: larger noise, smaller a
Drives	ELF $\approx$ 25 Hz mostly to I; INF $\approx$ 15 Hz modest to E/I (jittered)

Table 1: Fixed settings for the reported deterministic run.

3.3 Tier 3: Perceptual mapping

A monotonic illustrative mapping links condition-wise mean τ to false-positive rate (FP):

$$\mathrm{FP} = \sigma\big(\alpha + \beta\,(\tau_0 - \tau)\big), \quad \alpha = \mathrm{logit}(0.1),\ \beta = 12,\ \tau_0 = \text{baseline (A) mean},$$

with σ logistic; chosen to keep FP in a realistic 0.09–0.10x band while preserving order.

3.4 Tier 4: Spatial gradient

Assume an INF hotspot near $\sim$75 m and ELF variance decaying with distance; a smooth Gaussian-like profile is mapped through a logistic link to predict report probability vs. distance.

4 Results (exact deterministic outputs)

Tier 1: Coincidence score (preview rows)

coincidence_zsum
3.521
4.342
3.086
1.647
1.953
3.696
4.092
1.720
0.012
1.049
2.633
1.777

Figure 1: Tier 1: Standardized coincidence score $C = z(\mathrm{ELF}_{20\text{–}30}) + z(\mathrm{INF}_{10\text{–}20}) + z(\mathrm{GM})$; preview of 12 rows.

Condition	$\mathbb{E}[\tau]$	SD	Frac < baseline 10th
A (low arousal)	0.267	0.482	0.103
B (high arousal)	0.266	0.464	0.111
C (high+ELF)	0.263	0.461	0.154
D (high+INF)	0.278	0.440	0.043
E (high+ELF+INF)	0.268	0.449	0.103

Figure 2: Tier 2: Exact aggregates from 2 s windows (0.5 s stride). Baseline-decile computed from A.

$\mathbb{E}[\tau]$	FP rate
0.267	0.100
0.266	0.101
0.263	0.104
0.278	0.089
0.268	0.098

Figure 3: Tier 3: Exact mapping from Tier 2 means using the logistic. Lower $\tau \Rightarrow$ higher FP. Condition D shows reduced FP due to its slightly higher τ.

Distance (m)	Pred. prob.
25	0.100
50	0.173
75	0.220
100	0.173
150	0.067
200	0.060

Figure 4: Tier 4: Distance–probability pairs (synthetic proximity prediction); peak near $\sim$75 m.

Tier 2: Condition summary

Tier 3: Perceptual mapping

Tier 4: Spatial gradient

5 Interpretation and scope

Rare multi-driver co-peaks exist (Fig. 1). Under high arousal, small changes in synchrony/entropy can shift the τ-proxy (Fig. 2). Mapping τ to FP is monotone (Fig. 3). A plausible proximity profile peaks near an INF hotspot (Fig. 4). In this deterministic run, high+INF (D) slightly increased τ (and lowered FP), showing that the sign of ELF/INF effects is *parameter-sensitive* and must be empirically constrained. TSZ emphasizes falsifiable *structure* (co-occurrence, band specificity, spatial prediction) rather than a fixed sign.

Why TSZ helps. TSZ offers a testable account for *state-dependent inflation* of ambiguous reports—especially multi-witness clusters near infrastructure—without invoking exotic physics. ENT interprets "entities" claims as statements about τ-systems; TSZ remains agnostic on provenance (adversarial tech, atmospheric, artifacts). The framework focuses measurement on the human-perceptual channel.

6 Limitations, safeguards, and falsifiers

Synthetic data. Simulations show *possibility*, not proof. Current numbers are for seed `424242`; other seeds/parameters can flip directions.
Parameter sensitivity. ELF/INF may either increase or decrease τ; sign must be empirically constrained.
Physiology anchors. Magnetophosphene thresholds for 20–60 Hz ocular/occipital exposure are $\sim 10 - -50$ mT, far above typical public ELF levels (μT). TSZ models *bias*, not induction.
Falsification. TSZ is weakened or falsified if: (i) co-occurrence in plausible bands fails to predict reports; (ii) band specificity disappears; (iii) proximity predictions fail; or (iv) ionizing dose drives effects.

Ethics and competing interests

No human subjects were studied. The authors declare no competing interests.

Data availability

Exact numbers in Figures 1–4 come from a deterministic run (seed `424242`). A separate appendix file—*TSZ Simulations: Exact Numerical Evidence*—contains the same values as standalone tables in the ENT repo. Scripts and expanded runs will be released upon submission.

7 Next steps

(1) Preregister parameter sweeps (ELF/INF magnitudes, arousal, coupling). (2) Record site spectra (ELF, INF, geomagnetic) with reports and arousal proxies (HRV/EDA). (3) Constrain parameters to empirical distributions. (4) Retest τ shifts and FP mapping. (5) Evaluate co-occurrence and band specificity. (6) Test proximity gradients near infrastructure with blinded observers. (7) Publish confirmatory analyses.